Beyond the Complaint Count: How Toronto 311 Service Requests Have Changed Over Time

Milestone 4 – Final Results & Report

Kezia Mathew

CIND820 DAH - Big Data Analytics Project - P2026

Supervisor: Dr. Tamer Abdou
Toronto Metropolitan University

July 27, 2026

Table of Contents

Project Overview

This project analyzes Toronto 311 customer-initiated service request data to examine how reported municipal service demand has changed over time. The analysis compares two four-year periods: 2010–2013, representing the earliest full-year period available in the dataset, and 2022–2025, representing the most recent complete full-year period.

The project is written from the position of a municipal data analyst. This project supports how municipal staff can use historical and recent 311 request patterns to understand reported service demand while avoiding unsupported claims about actual community need. This distinction is important because 311 request volume may reflect not only service issues, but also changes in reporting behaviour, public awareness, access to 311, service categories, and data publication practices.

The analysis will focus on service request categories, city divisions, monthly and seasonal patterns, and broad reporting profile changes. Ward-level comparisons will be treated cautiously because ward structures changed between the early and recent periods and should be noted.

Changes Made

Two major changes were made based on feedback provided and troubleshooting:

Correction to data loading: I loaded the 2022 to 2025 files using on_bad_lines="skip" because pandas reported “Expected 9 fields, saw 10.” I read that error as meaning each file had one badly formatted row. While checking my row counts for Milestone 4, I found this was not correct. The error message only names the first line it fails on, not every line. When I counted the affected rows directly, 134,296 rows were being deleted across the four recent files, which is 7.5% of the recent period. The cause is a difference in file formatting between the two periods. The 2010 to 2013 files wrap their values in quotation marks, so a comma inside a value is handled correctly. The 2022 to 2025 files contain no quotation marks at all. This means any value containing a comma split into an extra column, and the row is discarded. The deleted rows were not a random sample. Almost all of them belonged to a single city division whose name contains a comma: Parks, Forestry & Recreation, which was renamed Environment, Climate & Forestry in 2025. It wiped an entire city division from my Milestone 3 analysis. I corrected this by repairing the affected rows instead of dropping them, joining the split Division value back together. All results in this report are based on the repaired data.

Year	Rows kept by skip	Rows after repair	Recovered
2022	407,369	442,950	35,581
2023	385,975	414,282	28,307
2024	402,824	437,043	34,219
2025	464,080	500,269	36,189
Total			134,296

Addition to analysis method: in Milestone 3, I wrote that I would not use regression or cross-validation as main methods. I changed this for two reasons. First, my own Milestone 3 result showed the limitation of the baseline. The seasonal average model missed the February and December spikes because it can only repeat the average of past months, with no way to represent a trend. That suggested testing a model that could. Second, Milestone 4 requires comparing results against a baseline and naming a final model with a justification, which is not possible with only one model. The descriptive comparison from Milestone 3 is unchanged; the model comparison is added on top of it. I also added a statistical testing section. In Milestone 3 I compared percentages by eye and described the differences in words. For Milestone 4 I tested those differences using methods covered in CMTH642, so that claims about change are supported by a test result and an effect size rather than by visual inspection alone.

Decision Context: Who Benefits and Who is it for?

This project is written from the perspective of a municipal data analyst supporting Toronto's service planning and operational decision-making. The main users of this analysis would be operations staff, public services workers, policy analysts, and city council members.

The analysis supports those who need to understand how consumer reported 311 service request patterns have changed over time. The project helps to identify whether the most common service request categories, city divisions, and seasonal patterns have shifted by comparing the earliest (2010-2013) and most recent (2022-2025) full-year periods. The analysis can understand where 311 requests are showing up repeatedly, which services may be getting more reports, and where staff may need to look more closely before making decisions. For example, if snow-related requests always rise in certain months, or if waste requests make up a larger share in the recent period, that does not automatically mean the city should move resources right away. It means the pattern is worth reviewing. This project does not decide where the city should put its staff, funding, or services. Instead, it helps decision-makers interpret the open 311 dataset more carefully by showing what the data can reveal, what it cannot prove, and where caution is needed because of changes in categories, divisions, wards, data coverage, or reporting behaviour.

Research Questions

Overall Research Question: How have Toronto 311 reported service request patterns changed between 2010–2013 and 2022–2025, and what limitations should be considered when using these patterns for municipal service planning?

RQ1: How has reported 311 demand concentration by service request category and city division changed between 2010–2013 and 2022–2025?

RQ2: Which high-volume 311 service request categories show different monthly or seasonal patterns between 2010–2013 and 2022–2025?

RQ3*:* Which differences between the early and recent periods can be compared responsibly, and which require caution because of changes in categories, divisions, wards, data coverage, or reporting practices?

Research Question Results

Research Question	Required Fields	Method Used	Result	Stakeholder Use
RQ1: How has reported 311 demand concentration by service request category and city division changed between 2010–2013 and 2022–2025?	Creation Date, Service Request Type, Division, Section, Status, request count	Combined the eight yearly files using a repaired loader. Created a Period variable. Converted Creation Date to datetime. Ranked categories and divisions by count, percentage share, and cumulative share. Tested the change with a chi-square test of independence and reported Cramér's V as the effect size.	990 unique service request types; the top 20 account for 35.07% of records. Solid Waste Management Services fell 44.64% → 33.93%, Municipal Licensing & Standards rose 13.12% → 24.54%, Transportation Services rose 15.98% → 23.79%. χ^2 = 384,422.38, df = 11, Cramér's V = 0.356 (medium effect).	Operations staff and service managers — shows whether reported workload has shifted across service categories and city divisions over time.
RQ2: Which high-volume 311 service request categories show different monthly or seasonal patterns between 2010–2013 and 2022–2025?	Creation Date, Service Request Type, Division, year, month, season, request count	Derived year, month, and season fields. Compared monthly counts and monthly shares between periods. Checked normality with Shapiro-Wilk, then used the Wilcoxon signed-rank test on volume and a chi-square goodness of fit test on seasonal shape. Modelled the forecast per division rather than on total volume only.	Volume rose in all 12 months without exception (exact p = 0.000488, rank-biserial r = 1.000). Seasonal shape shifted only slightly (Cohen's w = 0.112, small effect; mean share change 0.77 points). Best forecasting model differs by division: OLS for Municipal Licensing & Standards (WAPE 5.54%), seasonal historical-average for Toronto Water (8.51%).	Service managers and city divisions — identifies recurring and changing seasonal pressures per service area, and shows which areas are predictable.
RQ3: Which differences between the early and recent periods can be compared responsibly, and which require caution because of changes in categories, divisions, wards, data coverage, or reporting practices?	Service Request Type, Division, Ward, Status, unique value counts, record coverage, missing values, duplicates	Audited each field by comparing the values used in each period, measuring both label overlap and record coverage. Traced division renaming across the study period. Compared status vocabularies between periods.	Division is safe to compare (91.63% record coverage); Section requires caution (83.77%); Service Request Type (62.32%), Status (14.58%), and Ward (1.49%) are not comparable. Forestry appears under three different division labels, and that renaming caused a division-level forecast to fail. The status vocabulary was almost entirely replaced.	Policy analysts, councillors' offices, and municipal decision-makers — prevents misleading conclusions when comparing 311 request patterns across time.

Methodological Rational and Literature Review

The 311 Toronto service officially launched on September 24, 2009, with the purpose of giving residents and businesses one easy way to request information or submit a non-emergency service request to the city (Deloitte, 2013, p. 6). Since then, 311 has become an important part of how Toronto communicates with the public and organizes service delivery (Stowers, 2022, p. 14). Instead of residents having to figure out which City division or department to contact, 311 provides a centralized system where service requests can be submitted, tracked, and directed to the appropriate area. The main goal of this project is to examine how customer-initiated 311 service demand has changed over time by comparing the earliest complete period available for this project, 2010–2013, with the most recent complete period, 2022–2025. The dataset is available through the City of Toronto Open Data Portal in CSV format. Each row represents one customer-initiated service request, which makes the dataset useful for grouping and comparing records by time-period, service request type, city division, month, and season.

The literature supports using 311 data to study reported municipal service demand. Stowers (2022) uses real 311 service request data from open data portals across 29 cities and treats each service request as the unit of analysis. This is similar to my project because each row in the Toronto 311 dataset will also be treated as one reported service request. Stowers also uses descriptive statistics, service categories, and visualizations to compare 311 systems. This supports my use of counts, percentage shares, rankings, tables, and charts to identify which service areas appear most often in the data. Stowers (2022, p. 17) also shows why it is important to examine the types of service requests, not only the total number of requests. In that study, common categories included garbage and recycling, code enforcement, parking, bulk item pickup, and aband vehicles. This matters for my project because total 311 request volume alone does not explain what kinds of issues residents are reporting. To understand how reported demand has changed over time, I need to compare both the number of requests and the types of requests between 2010–2013 and 2022–2025. Research on 311 systems also shows that request volume can be influenced by resident behaviour, not only by the number of service problems in an area. Stowers (2022, p. 16) summarizes studies showing that residents may be more likely to report issues when they feel a sense of concern or "ownership" over their neighbourhood. Other research also suggests that 311 use can be shaped by technology use, satisfaction with city services, neighbourhood conditions, and the characteristics of the people living in an area. This is important for my project because 311 request counts should be interpreted as reported demand, not as a complete measure of actual municipal need.

Wang et al. (2017, pp. 4–5) supports looking at the structure of 311 service categories to identify patterns in the data. In their study, they created a 311 "pattern signature" by calculating the relative frequency of each service request type within an area. This means they did not only look at the total number of 311 requests but also examined what share of requests belonged to each category. This is useful for my project because I am also comparing the composition of 311 requests, not just the total volume. By using counts, percentage shares, rankings, and category comparisons, I can identify whether the types of reported 311 demands changed between 2010–2013 and 2022–2025. Wang et al. (2017, p. 5) used these category-based patterns to group areas with similar 311 request profiles through clustering. While my project does not rely on clustering as the main method, the same logic supports my planned analysis. The categories in the 311 datasets can help reveal patterns in how residents report municipal service issues. For my project, this means service request types can be used to compare early and recent reporting profiles and to identify which categories appear more common, less common, newly added, or difficult to compare over time. The Toronto Open Data guest post comments on how the analysis is affected by changing request categories, ward boundary changes, missing fields, and partial dataset coverage (Lal, 2026). For this reason, my project treats the findings as patterns in the available open 311 records, not as a complete measure of all 311 demand in Toronto.

White and Trump (2018) provide another important caution. They explain that 311 requests should not automatically be treated as a measure of civic participation or actual neighbourhood need. Request volume can be affected by many factors, including whether residents know about 311, whether they trust the service, whether they have time to report issues, and whether they face language or access barriers (White and Trump, 2018, p. 15-17). This supports one of the main limitations of my project. A high number of requests in one area does not automatically mean that area has worse service conditions. A low number of requests also does not mean there are fewer problems. It may simply mean fewer people are reporting issues through 311. This is especially important for any geographic analysis in my project. Since the dataset does not include demographic information about who submitted each request, I cannot make strong claims about resident behaviour, inequality, or actual service need. I can describe where reported requests appear in the data, but I need to be careful not to overstate what those patterns mean.

Based on the literature, my project starts with a simple descriptive baseline method and then adds more careful comparison steps. First, I will combine the yearly CSV files and label each record by year and period. Second, I will profile the dataset by checking column structure, missing values, duplicates, date coverage, service request types, division labels, ward values, and status values. Third, I will compare service request categories and divisions using counts, percentage shares, rankings, and cumulative shares. Fourth, I will create date-based fields such as year, month, and season to study time-based patterns. Finally, I will include a limitation and comparability review to explain which findings can be compared clearly and which ones need caution. Overall, the literature supports the main design of my project. 311 data is useful for studying reported municipal service demand, but it is not a perfect measure of actual need. For that reason, I will combine descriptive statistics, category comparisons, division summaries, seasonal analysis, data profiling, and limitation analysis rather than using just the plain data to determine demand and patterns.

Baseline Method and Gap Analysis

The baseline method for this project is simple frequency analysis. This means counting 311 service requests by service request type, city division, year, month, season, and comparison period. The baseline outputs would include frequency tables, top category rankings, top division rankings, bar charts, and basic percentage share tables.

This baseline is useful because it gives a clear first look at the dataset. It can show which request categories appear most often, which city divisions receive the most reported requests, and whether the total number of requests differs between 2010–2013 and 2022–2025. It is also easy to reproduce and easy for non-technical stakeholders to understand. For this reason, basic descriptive analysis is an appropriate starting point for this project.

However, the baseline method is limited because it mainly answers, “what appears most often?” It does not fully explain whether the early and recent periods are comparable, whether categories changed over time, or whether certain request types follow seasonal patterns. A simple top 10 ranking may show that one category has a higher count in the recent period, but it does not explain whether that change reflects a real shift in reported demand or a change in how the city labelled, grouped, or published the request type. The planned analysis improves on the baseline by adding more context to the counts. Instead of only reporting totals, the project will calculate percentage shares, rankings, and cumulative shares so that the early and recent periods can be compared more fairly. Monthly and seasonal variables will also be created from the Creation Date field to identify recurring patterns across the year. Finally, a comparability check will be used to flag issues such as changed category names, division labels, ward values, missing fields, and other differences that may affect interpretation.

The gap between the baseline and the planned method is important because this project is not just a “top 10 complaints” analysis. The goal is to understand how reported 311 demand patterns changed over time and how carefully those changes should be interpreted. The baseline shows the basic distribution of requests, while the planned method adds the context needed to compare the two periods more responsibly.

Analysis Method

This project used Python and pandas for all data preparation and analysis, following the approach used in the course labs. Statistical tests used SciPy, and the regression models used statsmodels.

The yearly CSV files for 2010–2013 and 2022–2025 were combined into one dataset, with a source year field and a comparison period field added so each record could be identified as belonging to either period. The Creation Date field was converted to datetime format, and year, month, month name, weekday, and season fields were derived from it. Season was assigned by month, with Winter as December through February, Spring as March through May, Summer as June through August, and Fall as September through November.

The main categorical fields (Service Request Type, Division, Section, Status, Ward, and the first three characters of the postal code) were checked for missing values, formatting differences, and label changes across years. Categories were not merged or renamed, because category instability is itself part of what RQ3 examines.

The main analysis used frequency counts, percentage shares, rankings, and cumulative shares, since almost all fields are categorical. Monthly and seasonal aggregates were used for the time-based analysis.

Dataset Biography

A major limitation of the Toronto 311 open dataset is that it does not represent the full universe of 311 activity. According to a Toronto Open Data guest post, the open 311 dataset represents only approximately 30–35% of total 311 requests and covers only 6 of the city's 45 divisions. This means the dataset is suitable for analyzing patterns in the open customer-initiated service request records, but it should not be treated as a complete measure of all 311 activity or all municipal service demand.

The geographic fields also require caution. The dataset includes the first three characters of the postal code, which represents a larger area rather than a specific postal code. This protects privacy by avoiding precise household-level location information, but it also limits the level of geographic analysis that can be performed. The dataset also includes ward information, but direct ward comparison over time is limited because Toronto's ward boundaries changed between the early and recent comparison periods. The data is published under the City of Toronto Open Data Licence, which permits use and redistribution with attribution. No licence condition restricts the analysis performed here.

Data Structure and Integration Plan

The Toronto 311 data is organized as yearly CSV files, with each row representing one reported service request. The dataset is already in a flat table format, so I do not need to join multiple tables together. The main integration step will be combining the yearly files into one larger dataset. The main columns include Creation Date, Status, First 3 Chars of Postal Code, Ward, Service Request Type, Division, Section, and intersection street fields. Most of these columns are text-based or categorical. The Creation Date column will be converted into a date format so I can create year, month, and season columns for the time-based analysis.

An important structural difference between the two periods was found during Milestone 4. The 2010–2013 files are quoted CSV files, while the 2022–2025 files are unquoted. This difference is not documented in the dataset description and is not visible from the column structure, but it determines whether values containing commas can be parsed correctly. This is described in the Changes Made section and in the data quality results below.

Data Results

The combined repaired datasets contains 3,027,264 records: 1,232,720 from 2010 to 2013 and 1,794,544 from 2022 to 2025.

Period	Year	Records
Early	2010	254,218
Early	2011	285,333
Early	2012	307,299
Early	2013	385,870
Recent	2022	442,950
Recent	2023	414,282
Recent	2024	437,043
Recent	2025	500,269

All Creation Date values converted successfully to datetime format, and there were 0 non-parseable dates. The date range runs from January 1, 2010 to December 31, 2025. The years 2014 to 2021 are not included in this project.

The data quality review showed that the main fields used in the analysis were complete. Creation Date, Status, Ward, Service Request Type, Division, Section, Source Year, and Period had no missing values. The main missingness issue was in the intersection street fields, which are missing about 87.7% of their values. Because of this, intersection-level analysis was not used as a main part of the project. This missingness makes sense because intersection details would make it easier to find where an individual report came from.

A duplicate check found 2,318 duplicate rows, which is about 0.08% of the full dataset. The dataset does not include a unique service request ID, so I cannot confirm whether these are true data errors or separate reports that happen to have the same values. I documented them as a data quality issue instead of removing them. At 0.08% they are too small to change the percentage shares and rankings this project uses.

I am reporting the parsing issue in full instead of quietly fixing it, because it shows a risk for anyone else using this dataset. The failure was silent. The code ran without any error, produced a dataset that looked normal, and gave results that looked reasonable. Nothing in the output showed that 7.5% of the recent period was missing. I only found it because I checked the row counts against the raw files. The rows that were deleted were the ones where a value contained a comma, so the loss was systematic and not random. This means it affected the division comparison specifically instead of adding noise evenly across the dataset.

Cleaning and Preparation

The cleaning and preparation step turned the audited yearly CSV files into one analysis-ready dataset. After the EDA confirmed that the files had the same main structure, the 2010, 2011, 2012, 2013, 2022, 2023, 2024, and 2025 files were combined into one dataset. The 2022 to 2025 files were loaded using the repaired function described in the Changes Made section rather than the default settings, because the default behaviour silently dropped rows. I added a source year field and a comparison period field so that each record could be identified as belonging to either 2010–2013 or 2022–2025.

The Creation Date field was converted into datetime format, and all values converted successfully with 0 non-parseable dates. From this field I created new columns for year, month, month name, weekday, and season. These variables were needed for the monthly and seasonal parts of the analysis. I also checked that the year taken from Creation Date matched the source year for every record, and all 3,027,264 rows matched.

The main categorical fields were cleaned and prepared for grouping. These include Service Request Type, Division, Section, Status, Ward, and the first three characters of the postal code. I removed extra spaces and converted blank values to missing values so that the missing value checks would count them correctly. I did not combine or rename any service request categories, even where the labels looked similar, because category changes are part of what RQ3 examines. Merging them would have hidden the exact differences the comparability audit was built to find.

Missing values were handled based on how important each field was to the research questions. Creation Date, Service Request Type, Division, Status, Ward, and Section had no missing values. The intersection street fields are missing about 87.7% of their values, so they were not used as main analysis fields. Duplicate records were checked before the final analysis because repeated rows could affect request counts, rankings, and percentage shares. The check found 2,318 duplicates, which is 0.08% of the dataset, and I documented them rather than removing them because there is no unique request ID to confirm whether they are true duplicates.

After cleaning, the final dataset kept the fields needed for the research questions and was used to produce the summary tables, charts, seasonal analysis, statistical tests, comparability audit, and forecasting model.

RQ1: Demand Concentration by Category and Division

Reported 311 demand is spread across many service request categories instead of being concentrated in a few. The dataset contains 990 unique service request types. The largest single type is Property Standards, but it only makes up 3.12% of all records. The top 20 types together make up 35.07% of records, and the top 10 make up 23.07%. This supports the use of rankings and percentage shares for RQ1, because no small group of categories dominates the dataset. The five largest service request types are Property Standards with 94,487 records or 3.12%, Residential Furniture / Not Picked Up with 83,812 or 2.77%, General Pruning with 81,628 or 2.70%, Sewer Service Line-Blocked with 71,653 or 2.37%, and Res / Garbage / Not Picked Up with 66,825 or 2.21%.

The highest-volume categories changed between the two periods. In the early period the leading types were Residential Furniture / Not Picked Up at 4.18% of early records, Property Standards at 4.00%, and Sewer Service Line-Blocked at 3.46%. In the recent period the leading types were Injured – Wildlife at 3.60%, Residential: Bin: Repair or Replace Lid at 3.14%, and General Pruning at 2.64%. Some categories appear in both periods, but their rankings and shares changed.

The division-level changes are larger.

Division	**Early period**	**Recent period**	**Change**
Solid Waste Management Services	44.64%	33.93%	−10.71
Municipal Licensing & Standards	13.12%	24.54%	+11.42
Transportation Services	15.98%	23.79%	+7.81
Toronto Water	15.76%	7.70%	−8.06
Urban Forestry	8.03%	0.00%	−8.03
Parks, Forestry & Recreation	0.00%	5.47%	+5.47
Environment, Climate & Forestry	0.00%	2.02%	+2.02
311	2.46%	1.67%	−0.79
Unknown	0.00%	0.81%	+0.81

Testing the division change

In Milestone 3 I described this change by comparing percentages. For Milestone 4 I tested it. My null hypothesis is that Division and Period are independent, which would mean the division mix did not change. My alternative hypothesis is that Division and Period are associated, which would mean the mix did change.

A chi-square test of independence needs all expected cell counts to be at least 5. I checked this and the smallest expected count was 18.3, so the assumption is met.
The chi-square statistic was 384,422.38 with 11 degrees of freedom, and the p-value was far below 0.001.

However, the dataset has 3,027,264 records, and with a sample that large almost any difference will come out significant. This means the p-value on its own does not tell me very much. So I also calculated Cramér's V, which measures how big the difference is instead of only whether it exists. Cramér's V was 0.356, which counts as a medium effect. This is the more useful result because it shows the division mix changed by a moderate amount, not by an amount that only looks meaningful because the dataset is large.

These results should be interpreted as changes in the reported 311 records, not as proof that actual municipal service need changed by the same amounts.

RQ2: Monthly and Seasonal Patterns

Both periods have higher reported request volume in the warmer months.

Season	Early period	Recent period
Spring	23.46%	25.08%
Summer	30.24%	28.59%
Fall	25.75%	23.57%
Winter	20.55%	22.76%

The recent period has many more records overall, so comparing monthly counts directly would mostly measure that growth instead of a change in seasonal timing. Because of this I tested two separate questions.

Choosing the test

A Shapiro-Wilk test on the monthly counts gave $p = 0.3847$ for the early period and $p = 0.9455$ for the recent period, so normality was not rejected for either one.

This is a paired comparison because each of the 12 calendar months appears in both periods, so the pairs match naturally. Since normality was not rejected, either a paired t-test or a Wilcoxon signed-rank test would be acceptable. I used the Wilcoxon signed-rank test as my main result because there are only 12 pairs, and with a sample that small the Shapiro-Wilk test is not very good at detecting non-normality. The non-parametric test is the safer choice. I also ran the paired t-test as a robustness check, and both tests agree.

Test 1: monthly volume

My null hypothesis is that the median difference in monthly volume between the periods is zero. My alternative hypothesis is that it is not zero. The Wilcoxon test gave $W = 0.0$ with an exact p-value of 0.000488 for 12 pairs. The paired t-test gave $t = -12.148$ with $p = 1.03 \times 10^{-7}$. For effect size, the matched-pairs rank-biserial correlation was $r = 1.000$. This is the largest value possible and it means all 12 months moved in the same direction. Cohen's dz was 3.507, which is a very large effect. This shows that reported demand grew across the whole year, not only in certain seasons.

Test 2: monthly share

Next I converted each month into a percentage of its own period total, which removes the volume difference. The Wilcoxon test gave a statistic of 38.0 with an exact p-value of 0.9697, and the paired t-test gave $t = 0.0094$ with $p = 0.9927$. These results need to be read carefully. The 12 monthly shares add up to 100% in each period, so the 12 differences between the periods have to add up to zero. A paired test checks whether the average difference is different from zero, but here the average difference is zero because of how percentages work, not because of anything in the data. The paired t-test makes this obvious, because a t-statistic of 0.0094 is almost exactly zero and it could not have been anything else. Neither test is wrong, but neither one can answer the question I actually wanted to ask.

Test 3: chi-square goodness of fit

The right test for this question is a chi-square goodness of fit test, which checks whether the recent monthly counts match what the early period's monthly pattern would predict.
My null hypothesis is that the recent monthly distribution follows the early-period monthly pattern. My alternative hypothesis is that it does not.
The assumption is that all expected counts are at least 5. The smallest expected count was 114,071.9, so this is met. The test gave a statistic of 22,530.4 with 11 degrees of freedom, which is significant well below 0.001. As with the division test, the large sample size makes significance almost certain, so I calculated an effect size as well. Cohen's w was 0.112, which is a small effect.

Test 4: association between the two monthly patterns

In Milestone 3 I described the two monthly patterns as similar without testing it. For Milestone 4 I tested it with a named test. My null hypothesis is that there is no monotonic association between the early and recent monthly patterns, meaning rho = 0. My alternative hypothesis is that there is one. Spearman's rank correlation only needs ordinal data and a monotonic relationship, and both are met across the 12 paired months. I used Spearman instead of Pearson because the question is about the order of the months by volume, not about a straight-line relationship in the raw counts. Spearman's rho was 0.6364, and I also calculated Pearson's r as 0.7323 for comparison. Rho is the effect size itself. Rho squared is 0.405, which means about 41% of the rank variance is shared. This is a strong association.

What these results mean for RQ2

Putting all four tests together gives a clearer answer than any one of them alone.
Reported 311 demand grew a lot, and it grew in every single month with no exceptions. The seasonal shape also changed, but only by a small amount. Every monthly share moved by less than two percentage points. The largest changes were February at +1.76 and January at +1.25, and the average absolute change across all 12 months was 0.77 percentage points.
The direction of the change is consistent. January, February, and May gained share, while July through December all lost share. This matches the season table, where Winter went from 20.55% to 22.76% and Summer went from 30.24% to 28.59%. In simple terms, the summer peak has softened and reported activity is spread a little more evenly across the year than it was a decade ago.

The heatmap of the top 10 service request types adds something important. Even though the overall seasonal shape is fairly stable, the individual categories do not all follow the same pattern. Road – Pot hole peaks around March and April, while Injured – Wildlife and Property Standards are higher in the warmer months. So the overall stability should not be read as meaning every service area has the same timing.

RQ3: What Can Be Compared Responsibly

RQ3 asks which differences between the periods can be compared responsibly and which need caution. To answer this I audited each key field by comparing the values used in the early period against the values used in the recent period.

I measured the overlap in two different ways, and the difference between them matters. Label overlap is the share of category names that appear in both periods. Record coverage is the share of actual records that fall into categories present in both periods. These two numbers can be very different. Division has a label overlap of only 41.67%, which looks bad, but a record coverage of 91.63%, because the five divisions that appear in both periods hold almost all of the records. Coverage is the better measure of whether a comparison is safe, because it shows how much of the data is affected instead of how many rare labels exist.

Field	Shared	Label overlap	Early cov.	Recent cov.	Verdict
Division	5	41.67%	91.96%	91.63%	Safe
Section	12	42.86%	89.51%	83.77%	Caution
Request Type	348	35.15%	86.31%	62.32%	Not comparable
Status	3	33.33%	98.98%	14.58%	Not comparable
Ward	2	2.90%	1.49%	3.13%	Not comparable

Division

Over 91% of records in both periods fall into divisions that appear in both, so the division comparison I used for RQ1 is reliable. There is one exception, which is forestry.

Forestry: same work under three different names

Urban Forestry makes up 8.03% of the early period and then disappears completely from the recent period. It did not stop receiving requests. The work was reorganised into Parks, Forestry

& Recreation, which appears from 2022 to 2024 at 5.47% of the recent period, and was then renamed again to Environment, Climate & Forestry in 2025 at 2.02%.

This means the same city function shows up under three different division names inside one study period. If someone read the data without checking, it would look like Toronto stopped receiving forestry requests in 2022. That would be completely wrong. This is the clearest example in the dataset of why division labels cannot be treated as stable categories over time.

This is not only a caution. When I tried to forecast Parks, Forestry & Recreation as its own division, the model failed, because the 2025 test set was empty under that label. I had to combine all three forestry labels into one series before I could model the division at all. So the comparability problem actually broke a later step in the project, instead of only being a note in a table.

Status

The status values were almost completely replaced between the periods. The early period uses Cancelled, Closed, In-Progress, In-progress, Initiated, and Unknown. The recent period uses Cancelled, Closed, Completed, In Progress, New, and Unknown. Only Cancelled, Closed, and Unknown appear in both.

Completed is about 45% of the recent records and has no early-period equivalent, and Initiated has no recent equivalent. Only 14.58% of recent records are in a status that also exists in the early period, so status cannot be compared across the periods. This also explains the three spellings of "in progress" I noticed in Milestone 3. Two of them belong to the early period and one belongs to the recent period, so it is a difference between periods and not just messy data entry.

Service Request Type and Ward

Only 62.32% of recent records fall into a request type that also exists in the early period. The high-volume categories that appear in both periods can be compared directly, but a category that only appears in the recent period does not prove new demand, because it might be a renamed or newly split version of an older category.

For Ward there are 45 values in the early period and 26 in the recent period, and only 2 are shared. This is because of Toronto's ward restructuring, and it confirms the decision not to compare wards across periods. No ward comparison appears anywhere in this project.

The useful part of this audit is that it tells a reader which of my own findings to trust. The division and monthly findings use fields with high record coverage. Any claim about status, wards, or newly appearing request categories would not be supported, and I have not made those claims.

Model Comparison and Final Model Selection

Milestone 3 used one forecasting model. Based on the Milestone 3 feedback, this section names that model properly, adds a standard benchmark and two regression models, tests all four using rolling-origin backtesting, adds WAPE to the error metrics, and extends the analysis from total monthly volume down to the division level so it lines up with RQ2.

The four models

Same-month seasonal historical-average baseline. This is my Milestone 3 model. It predicts each month using the average of that same calendar month across all the training years. It is a descriptive baseline, not a time-series or machine learning model, so I am naming it that way. Seasonal-naive benchmark. This is the standard forecasting benchmark. It predicts each month using the value of that same month in the year right before it, and nothing else. It is different from the historical average, which uses all training years.

OLS trend and season. A linear regression using a year trend term plus month indicators. This is the multiple linear regression method from CMTH642 Module 6 and CIND123 Module 10. Poisson GLM. The same formula fitted as a generalized linear model with a log link. Monthly request counts are counts of events in a fixed time period, which is how CIND123 Module 6 defines a Poisson variable, and CMTH642 Module 8 taught that generalized linear models extend regression to non-normal responses using a link function.

Rolling-origin backtesting

I tested each model using an expanding window. For each origin, the model trains only on the months before the test year and then forecasts all 12 months of that year. Regular k-fold cross-validation is not appropriate here, because shuffling the data would let the model train on future months to predict past ones.

Model	Test year	Train n	MAE	RMSE	MAPE	WAPE
Historical-average	2023	12	4,925.00	5,476.35	15.25	14.27
Seasonal-naive	2023	12	4,925.00	5,476.35	15.25	14.27
Historical-average	2024	24	5,069.83	5,941.02	14.96	13.92
Seasonal-naive	2024	24	5,939.58	6,604.06	17.49	16.31
OLS trend + season	2024	24	6,733.00	7,291.79	18.38	18.49
Poisson GLM	2024	24	6,623.96	7,215.22	17.97	18.19
Historical-average	2025	36	6,461.89	8,630.23	14.86	15.50
Seasonal-naive	2025	36	5,444.50	9,656.26	12.22	13.06
OLS trend + season	2025	36	6,872.10	8,965.00	15.81	16.48
Poisson GLM	2025	36	6,867.07	8,939.06	15.79	16.47

At the 2023 origin the two benchmarks give exactly the same result, and this is not a coincidence. With only one training year available, the average of a given month is the same as that month's single value. The two regression models are not included at that origin because training would only give 12 monthly observations for 13 parameters, which makes them unidentifiable.
I added WAPE because MAPE gives every month the same weight no matter how many requests it had, which can make errors in low-volume months look worse than they are. WAPE takes the total absolute error as a share of total actual volume, so it gives a fairer relative comparison when monthly volumes are uneven.

Stability across origins

Model	MAPE mean	MAPE sd	WAPE mean	WAPE sd
Seasonal-naive benchmark	14.86	3.73	14.68	2.30
Seasonal historical-average	14.91	0.07	14.71	1.12
Poisson GLM	16.88	1.54	17.33	1.22
OLS trend + season	17.10	1.82	17.48	1.42

The seasonal-naive benchmark has a slightly lower average error than the historical average, 14.86% against 14.91% MAPE, but its standard deviation across origins is more than fifty times bigger, 3.73 against 0.07. It won the 2025 origin clearly and lost the 2024 origin badly. This happens because it depends entirely on one preceding year, so it copies whatever was unusual about that year.

I selected the same-month seasonal historical-average baseline as my final model for total monthly volume. The two benchmarks are basically tied on average error, so the decision comes down to stability instead of accuracy. The historical average gave consistent results at every origin I tested and the seasonal-naive benchmark did not. A model whose error moves by several percentage points depending on which year it is used on would be hard to rely on for planning.

Neither regression model earned its place at the total-volume level. The fitted year coefficient is −246 requests per year, and against monthly counts averaging about 37,000 that is basically flat, so the trend term adds variation without adding information. The OLS model has an R-squared of 0.676 but an adjusted R-squared of 0.507. That gap of almost 17 points shows that a lot of the apparent fit comes from having 13 parameters for 36 observations. The Poisson dispersion statistic is 578.67, where a value near 1 would mean the Poisson assumption of equal mean and variance holds. The counts are badly overdispersed, so the Poisson standard errors are not reliable and I do not interpret them. A Negative Binomial model would be the correct next step and I have noted it as future work.

Forecasting by Division

The Milestone 3 feedback pointed out that forecasting only total monthly volume does not line up with RQ2, since RQ2 asks about high-volume categories. So I repeated the comparison at the division level, training on 2022 to 2024 and testing on 2025.

Division	Best model (lowest WAPE)	WAPE	MAPE
Municipal Licensing & Standards	OLS trend + season	5.54%	6.93%
Toronto Water	Seasonal historical-average	8.51%	8.42%
Solid Waste Management Services	Seasonal-naive benchmark	9.47%	9.67%
Forestry (all three labels combined)	Seasonal-naive benchmark	17.11%	33.34%
Transportation Services	Seasonal historical-average	34.19%	27.77%

Three things come out of this, and none of them were visible when I only looked at total volume. No single model is best for every division. OLS gives the best forecast for Municipal Licensing & Standards, the historical average is best for Toronto Water and Transportation Services, and the seasonal-naive benchmark is best for Solid Waste Management Services and forestry. Picking one model for total volume hides this completely, which is the main reason the feedback asked for category-level modelling.

The trend model wins where there actually is a trend. Municipal Licensing & Standards almost doubled its share of reported requests between the two periods, so a year trend term has something real to work with. It gives a WAPE of 5.54%, which is the most accurate forecast anywhere in this project. This shows that the regression models were worth including even though they lost on total volume, and that the earlier result was about the aggregate series rather than about the models themselves.

Transportation Services cannot really be forecast with these methods. Its best WAPE is 34.19%, and its RMSE is about double its MAE, which means a few extreme months are driving most of the error. This fits with weather-driven requests like potholes and snow clearing, which a seasonal model cannot predict. I am reporting this as a real negative result instead of leaving it out.

Forecasting 2026 and 2027

The feedback also asked me to extend the forecast past the evaluation period. The table below comes from training on all of 2022 to 2025 and forecasting forward. I did not use any 2026 or 2027 actual values at any point.

Model	2026 total	2027 total
Seasonal historical-average	448,635	448,635
Seasonal-naive benchmark	500,269	500,269
OLS trend + season	497,316	516,790

For comparison, the actual annual totals were 442,950 in 2022, 414,282 in 2023, 437,043 in 2024, and 500,269 in 2025.

Two things about these forecasts are worth pointing out. The historical-average and seasonal-naive models give the exact same numbers for both years, because neither one has any way for a forecast to change over time. Only the OLS model gives different numbers for 2026 and 2027, because only it has a trend term. The seasonal-naive benchmark just repeats 2025, which means it carries the unusual February 2025 spike forward into both future years. That is a clear weakness of that benchmark.

This is a forecast and not an evaluation. I cannot measure forecast accuracy for 2026 and 2027 until those years are finished and published. These numbers are here to show how each model behaves in a real forward-looking situation, and they should not be read as validated predictions.

Compute cost

Model	Fit and predict time
OLS trend + season	0.039 ms
Seasonal historical-average	0.113 ms
Poisson GLM	0.819 ms

All of the models fit and predict in well under a millisecond on 36 observations, so speed did not affect my choice of final model. OLS is the fastest because it has a closed-form solution, and the Poisson model is the slowest because it is fitted iteratively. These differences would only start to matter at a much larger scale, like daily counts across all divisions.

On the repaired data the final model gives MAE 6,461.92, RMSE 8,630.28, MAPE 14.86%, and WAPE 15.50% on the 2025 test year. The largest single error is February 2025, where the actual count of 53,707 was 42.91% higher than the predicted 30,663. December has a similar problem. These two months are the main reason RMSE is so much higher than MAE, because RMSE punishes large errors more. A seasonal average model cannot predict unusual months by design, and this is a real limitation of the model I selected.

Comparison with Prior Work

Lal (2026) forecast Toronto 311 service requests at the ward level using the same open dataset. The results are not directly comparable, because that project reports variance explained for its best-performing division while I report MAE, RMSE, MAPE and WAPE, and because that model included weather features that mine does not. However, several of the findings align closely. Lal reports that the Environment division "had undergone multiple name changes from Parks to Urban Forestry to Environment," which independently confirms the renaming sequence my comparability audit identified. Lal also reports that forecasting performance varies by division and that transportation requests are the most difficult to predict because of irregular demand. My division-level results show the same pattern, with WAPE ranging from 5.54% for Municipal Licensing & Standards to 34.19% for Transportation Services.

One methodological difference is worth noting. Lal consolidated more than 200 request types into standardised categories to make long-term analysis possible. I deliberately did not merge categories, because category instability is what RQ3 examines, and merging would have concealed the differences the comparability audit was built to measure. The two approaches suit different goals: consolidation supports forecasting, while preserving the original labels supports assessing whether comparisons over time are valid at all.

Lal also found that extreme weather produced request volumes three to ten times higher than normal, including a single day in May 2022 that generated 1,575 environmental requests. This offers a likely explanation for the largest error in my own model, February 2025, where actual volume exceeded the prediction by 42.91%. A model with no weather inputs cannot anticipate months of that kind.

Answers to the Research Questions

RQ1: Has demand concentration by category and division changed?

Yes, by a moderate amount. Solid Waste Management Services went from 44.64% to 33.93% of reported requests, Municipal Licensing & Standards went from 13.12% to 24.54%, and Transportation Services went from 15.98% to 23.79%. Demand is still spread across many categories rather than concentrated. There are 990 different service request types and the top 20 only make up 35.07% of records.

Evidence: the division share table, the chi-square test of independence at 384,422.38 with 11 degrees of freedom and Cramér's V of 0.356 which is a medium effect, and the Pareto cumulative share analysis.

RQ2: Do monthly or seasonal patterns differ between the periods?

Reported volume grew in all 12 months with no exceptions. The seasonal shape also changed, but only slightly. The summer peak softened and winter gained share, and every monthly share moved by less than two percentage points with an average change of 0.77 points. At the category level the forecast results are very different from one division to another, which confirms that high-volume categories do not all follow the same seasonal pattern.

Evidence: the Wilcoxon signed-rank test on monthly volume with exact $p = 0.00049$ and rank-biserial $r = 1.000$, the chi-square goodness of fit test at 22,530.4 with 11 degrees of freedom and Cohen's w of 0.112, and the division-level forecast table where WAPE ranges from 5.54% to 34.19%.

RQ3: Which differences can be compared responsibly?

Division comparisons are safe, with 91.6% record coverage in both periods. Section comparisons need caution at 83.8%. Service Request Type, Status, and Ward are not comparable. Forestry has to be left out of any trend claim, because the same city function was renamed twice inside the study period, and that renaming caused a division-level forecast to fail.
Evidence: the comparability audit table showing both label overlap and record coverage, the Urban Forestry to Parks, Forestry & Recreation to Environment, Climate & Forestry sequence, the almost complete replacement of the status values, and 45 early ward values against 26 recent ward values with only 2 shared.

What the Results Mean for the People This Project Is For

For operations staff and service managers. The division mix of reported demand shifted by a moderate amount between the two periods. Municipal Licensing & Standards and Transportation Services now make up a much larger share of reported requests than they did a decade ago, and Solid Waste Management Services makes up a smaller share. This is worth reviewing, but it is not a staffing decision. The shift describes what residents reported through the open 311 channel, which is not the same as total workload.

For anyone using this dataset for planning. The comparability audit is probably the most useful output of this project, because it shows which fields can be compared over time and which cannot. It would stop an analyst from concluding that forestry demand collapsed in 2022, or that completion rates changed, when both of those are caused by renaming instead of by real change.

On forecasting. Total monthly volume can be predicted to about 15% error using a simple seasonal average, but the accuracy varies a lot by division, from 5.54% for Municipal Licensing & Standards to 34.19% for Transportation Services. So any forecasting use should be checked division by division instead of assuming the overall accuracy applies everywhere. The model also missed February 2025 by 42.91%, so it should be treated as a planning aid for normal months and not as something to rely on for resourcing decisions.

On data practice. The parsing issue in this report is a practical warning for anyone reusing this dataset. Parsed row counts should be checked against the raw files, because the default behaviour fails silently and the data it loses is systematic rather than random

Pipeline Diagram and Sequence of Operations

The pipeline for this project shows the order of steps from the raw yearly CSV files to the final analysis outputs. The purpose of the pipeline is to make the project easier to follow and easier to reproduce.

Figure 1.1

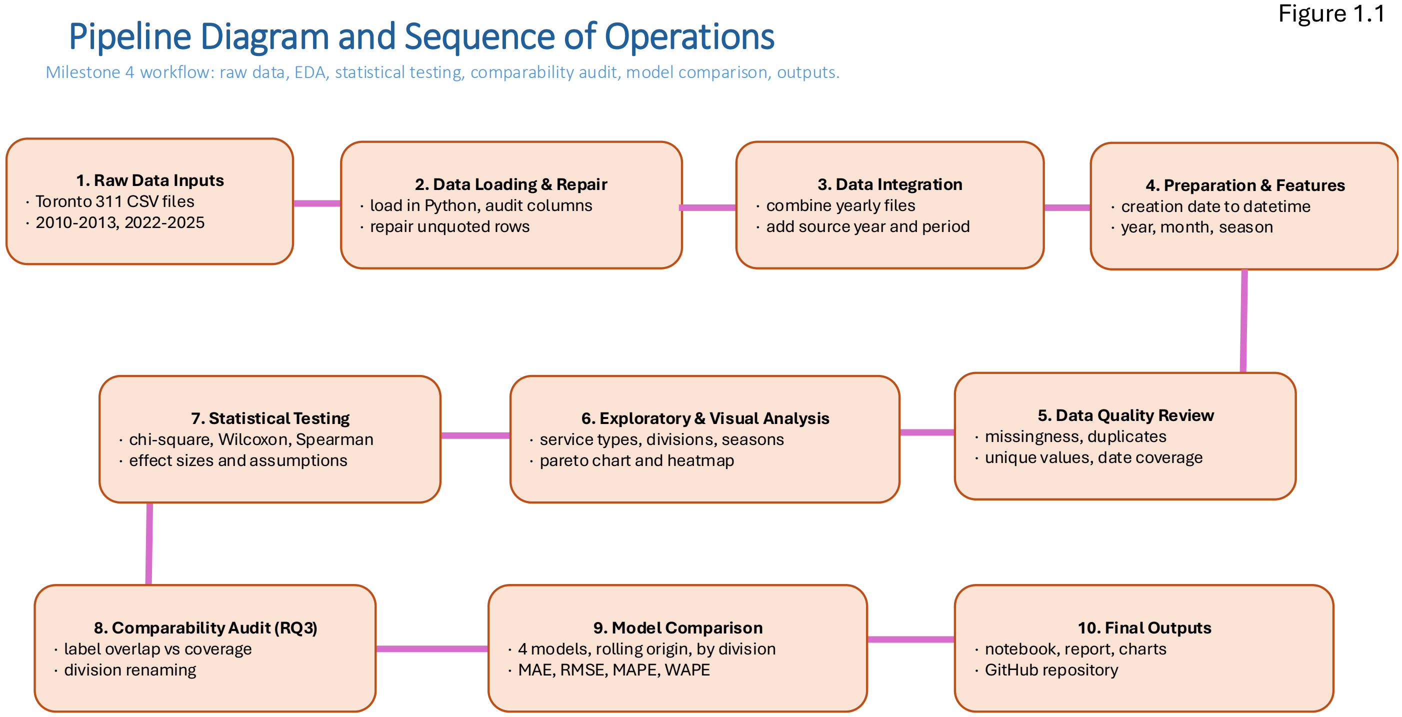

Figure 1.1 shows the project workflow from the raw Toronto 311 yearly CSV files to the final project outputs. The process begins with data loading and file auditing, followed by data integration, preparation, and feature engineering. After data quality review, the project moves

into exploratory analysis, visual analysis, and a simple baseline monthly prediction model. The final stages include model evaluation, interpretation of limitations, and final outputs such as the notebook, report, charts, GitHub repository, and video walkthrough.

Limitations: Association, Not Causation

This project can show patterns in Toronto 311 service request data, but it cannot prove what caused those patterns. If one request type increased between 2010–2013 and 2022–2025, the data can show that the increase appears in the available records, but it cannot prove that actual municipal need increased by the same amount.

A major limitation is that 311 data reflects 'customer' (emphasis on customer because it's consumer based and so it comes with bias on what they deem as a concern or what they feel they should be requesting) reported demand, not all community need. A higher number of requests may mean there are more service issues, but it may also reflect higher awareness of 311, easier access to reporting tools, repeated reports, or greater willingness to contact the City. A lower number of requests also does not automatically mean fewer problems, because some residents may face barriers such as language, time, technology access, or lower trust in government.

The articles I reviewed also point out that 311 data can be hard to interpret. White and Trump warn that 311 data usually does not show who made the request, and a small number of frequent users may affect request totals. Wang et al. and Stowers also show that service request categories can change over time. This matters for my project because service request labels, division names, section labels, and ward values may not be fully comparable between 2010–2013 and 2022–2025.

There are also limits with the Toronto dataset itself. The open 311 dataset only represents part of total 311 activity, and some fields have missing values or limited detail. The data can show that a request was made, but it does not fully explain the severity of the issue, the root cause, or whether the resident was satisfied with the result. These points are to be kept in mind when working with the data and not seeing the results as an absolute truth or the answer to forecasting requests. There are limitations and possible bias! There is a silent data loss risk in the source files. The 2022 to 2025 files are unquoted, so default parsing either drops or misaligns any row where a value contains a comma. In this project that removed 7.5% of the recent period without raising any error.Category and division labels are not stable over time. Division names, section names, and status values all changed between the periods, and in some cases the same work is recorded under a new name. Any category that appears or disappears should be checked for renaming before it is treated as a change in demand.

The validation depth is limited. I had three rolling origins for the two seasonal benchmarks and only two for the regression models, because the open dataset gives four recent years and the earliest origin does not leave enough observations to fit 13 parameters. This is enough to show whether performance is consistent, but not enough to make a strong general claim about stability. The Poisson model is misspecified for this data. The dispersion statistic is 578.67, so the assumption of equal mean and variance is badly violated and the model's standard errors should not be used.Duplicates were kept. The 2,318 duplicate rows are documented rather than removed, because there is no unique request ID to confirm whether they are errors or genuinely repeated reports.

Privacy, Misuse, Bias, and AI Use

This project uses data from the City of Toronto's 311 Service Requests dataset, published under the City of Toronto Open Data Licence. The dataset does not include any personal or identifying information such as names, phone numbers, email addresses, or full addresses. Geographic information is limited to broader fields such as ward, intersection, and the first three characters of the postal code, which reduces the risk of identifying individual residents or households.

The main privacy risk in this project is not direct identification but over-interpreting local service request patterns. Because of this, all results were reported at a broad level: service request type, division, comparison period, month, and season. No results were reported at ward or postal-code level. This decision was reinforced by the comparability audit, which found that only 2 of the ward values appear in both periods, so ward comparisons would not have been valid even if I had wanted to make them.

The main misuse risk is that these findings could be used to justify moving municipal resources based on reported request volume alone. Because request volume reflects who reports as well as what needs fixing, decisions driven only by 311 counts could end up underserving communities that face barriers to reporting. This report should be used to identify areas that are worth reviewing, not to rank areas by need.

Bias is a key limitation throughout this project because 311 data only shows reported requests. A high number of requests does not automatically mean greater actual need, and a low number of requests does not mean there are fewer problems. Request volume may reflect awareness of 311, access to reporting channels, language barriers, trust in government, or repeated reports from the same person. The open dataset also represents only part of total 311 activity, so all findings in this report are described as patterns in reported demand and not as proof of actual municipal need.

For Milestone 4 I used AI assistance (Claude) to review my notebook and report against the Milestone 4 requirements and my supervisor's feedback and identify what exactly I was missing and to help explain code for the statistical tests, the comparability audit, and the model comparison, including why each test was appropriate for my data I was not as familiar or needed a refresher with and to fix some of the outputs visually. AI was NOT used to determine results or do my coding work or report. AI-assistance was compared against the dataset, course instructions, uploaded articles, and city of Toronto sources. I understand that AI can produce incorrect code, fake citations, or unsupported interpretations, so the analysis and report are and will be my own work!

Github Repository and Reproducibility

A GitHub repository was created to organize the project files and support reproducibility. The repository contains the Milestone 4 Python notebook with all outputs saved, the compiled report, the figures used in the report, and a README file. It is organized into folders separating the notebook, the report, and the figures.

The README explains the project, lists the three research questions, links to the City of Toronto Open Data 311 dataset, and names the eight yearly CSV files required. It documents the folder the notebook expects those files in, the Python version and package versions used, and the steps to reproduce the analysis: download the CSV files, place them in the data folder, and run the notebook from top to bottom.

The README also includes a specific warning about loading the data. The 2022 to 2025 files must be loaded using the repair function included in the notebook rather than the default pandas settings with on_bad_lins="skip". Anyone who uses the default settings will silently lose 134,296 rows and obtain a materially different division comparison, so this step is documented explicitly rather than left for a user to discover.

For this project, reproducibility means another person should be able to see what data was used, what steps were followed, and how the results were produced. The README lists the expected record totals at each stage = 3,027,264 records overall, 1,232,720 in the early period and 1,794,544 in the recent period, so that a user can confirm their run matches mine.

References

Ahmad, L., Eng, J., Garofalo, N., Jappert, C., Nangal, V., Nathoo, A. N., Salem, A., Vukovic, S., Wright, S., & Lucas, J. (2025). *Do 311 service request systems increase responsiveness? Results of a field experiment*. https://www.researchgate.net/publication/391839881_Do_311_Service_Request_Systems_Increase_Responsiveness_Results_of_a_Field_Experiment

Canada NewsWire. (2009, September 24). *Toronto launches 311 service for easy access to your City*. https://ezproxy.lib.torontomu.ca/login?url=https://www.proquest.com/wire-feeds/toronto-launches-311-service-easy-access-your/docview/455159982/se-2

City of Toronto. (n.d.). *311 service requests – customer initiated*. City of Toronto Open Data Portal. Retrieved May 8, 2026, from https://open.toronto.ca/dataset/311-service-requests-customer-initiated/

City of Toronto. (n.d.). *311 service standard dashboard*. Retrieved May 10, 2026, from https://www.toronto.ca/home/311-toronto-at-your-service/311-frequently-asked-questions/311-service-standard-dashboard/

City of Toronto. (n.d.). *311 Toronto*. Retrieved May 10, 2026, from https://www.toronto.ca/home/311-toronto-at-your-service/

City of Toronto. (n.d.). *Open data*. Retrieved May 8, 2026, from https://www.toronto.ca/city-government/data-research-maps/open-data/

City of Toronto. (n.d.). *Open data licence*. Retrieved May 8, 2026, from https://www.toronto.ca/city-government/data-research-maps/open-data/open-data-licence/

City of Toronto. (2026, May 1). *Using 311 and other data to identify trends and improve service delivery in Parks and Recreation, Transportation Services and Municipal Licensing and Standards*. https://www.toronto.ca/legdocs/mmis/2026/se/bgrd/backgroundfile-286650.pdf

Deloitte. (2013). *Service efficiency study of the 311 Toronto Division: Final report*. City of Toronto. https://www.toronto.ca/wp-content/uploads/2017/08/969b-311-SES.pdf

Hashemi, M. (2022). Automatic type detection of 311 service requests based on customer provided descriptions. *Applied Artificial Intelligence, 36*(1), Article 2073717. https://doi.org/10.1080/08839514.2022.2073717

Lal, R. (2026, January 15). City of Toronto Open Data. *Guest post: Forecasting 311 service requests to support a more proactive Toronto*. https://open.toronto.ca/guest-post-forecasting-311-service-requests-to-support-a-more-proactive-toronto/

Stowers, G. N. L. (2022). Back to basics: City services and 311 service requests. *State and Local Government Review, 54*(1), 13–31. https://doi.org/10.1177/0160323X211064253

Wang, L., Qian, C., Kats, P., Kontokosta, C., & Sobolevsky, S. (2017). Structure of 311 service requests as a signature of urban location. *PLOS ONE, 12*(10), Article e0186314. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0186314

White, A., & Trump, K.-S. (2016). *The promises and pitfalls of 311 data*. https://dspace.mit.edu/entities/publication/00aeb97e-e25e-4fda-a761-c27c202f213e